# Extent of Grenvillian remanence components in rocks of the Southern Province

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Early Aphebian Gowganda sedimentary rocks and intruding Nipissing diabase sills were sampled for paleomagnetic study at 88 sites (~500 cores, ~1000 specimens) along two ~42 km long profiles extending north from the Grenville Front into the Cobalt Plate of the Southern Structural Province in the River Valley – Lake Temagami area of Ontario. After AF demagnetization a postfolding pre-Nipissing ~2200 Ma remanence was found in eight of the 37 Gowganda sediment sites that were >2 km north of the front, giving a pole at 109°W, 63°N ($d_p = 10°$, $d_m = 19°$). The Nippissing diabase from >2 km north of the front retains a stable antiparallel prefolding N1 remanence direction in 22 of 40 sites, giving a pole position of 85°W, 17°S ($d_p = 6°$, $d_m = 10°$). These "south and down" remanence directions found in the southern portion of the plate contrast with the antiparallel "north and up" directions found in the northern portion, thereby indicating the occurrence of either two nearly cogenetic Nipissing intrusive events or the sequential emplacement of the Nipissing during an Earth's magnetic field reversal across the plate. At two sites a Nipissing remagnetized remanence was found in Gowganda sediments with a pole of 115°W, 18°S. Also three "Nipissing" sites give a pole at 164°W, 3°N, which is close to the known pole for the later ~1.25 Ga Sudbury olivine diabase dikes. One site is adjacent to a large dike and two were found on thin-section examination to be olivine diabase. The eight sites in Gowganda sediment matrix and conglomerate clasts and in Nipissing diabase from within <2 km from the front were found to have a postfolding metamorphic remanence with a Grenville orogenic pole at 45°W, 51°N ($d_p = 19°$, $d_m = 21°$). Finally, the results lead to a suggested revision in the APW path for the ~2300–~1650 Ma interval for North America.

Des échantillons de roches sédimentaires de la formation de Gowganda de l'Aphébien inférieur et des intrusions en filons-couches de la diabase de Nipissing ont été prélevés pour une étude paléomagnétique sur 88 sites (~500 carottes, ~1000 spécimens) disposés selon deux profils de 42 km de long s'étalant en direction nord du Front du Grenville dans la Plaque Cobalt de la province tectonique du Sud dans la région River Valley – Lake Temagami en Ontario. Après avoir désaimanté dans un champ alternatif, on a trouvé une rémanence ~2200 Ma postérieure au plissement et antérieure à l'intrusion Nipissing dans huit des 37 sites de sédiments Gowganda situés >2 km au nord du front et fournissant un pôle à 109°O, 63°N, ($d_p = 10°$, $d_m = 19°$). La diabase de Nipissing à >2 km au nord du front conserve une direction de rémanence N1 stable antiparallèle antérieure à la phase de plissement dans 22 des 40 sites et fournissant une position du pôle à 85°O, 17°S ($d_p = 6°$, $d_m = 10°$). Ces directions de rémanence "sud et en bas" observées dans la partie sud de la plaque contrastent avec les directions antiparallèles "nord et sud" de la partie nord, indiquant ainsi l'apparition soit de deux phases d'intrusions presque cogénétiques de la diabase de Nipissing ou d'une mise en place séquencielle de cette diabase durant la période de renversement du champ magnétique de la Terre à la grandeur de la plaque. À deux sites une rémanence réaimantée de la diabase de Nipissing a été observée dans les sédiments de Gowganda avec un pôle à 115°O, 18°S. Aussi trois sites dans la diabase de Nipissing fournissent un pôle à 164°O, 3°N voisin du pôle connu pour les dykes tardifs de diabase à olivine de Sudbury ~1,25 Ga. Un site est adjacent à un grand dyke et pour deux autres un examen en lame mince a montré une composition de diabase à olivine. Les huit sites dans la matrice et les fragments conglomératiques des sédiments de Gowganda et dans la diabase de Nipissing à l'intérieur de <2 km du front ont révélé une rémanence liée à un métamorphisme postérieur à la phase de plissement avec un pôle orogénique grenvillien à 45°O, 51°N ($d_p = 19°$, $d_m = 21°$). Finalement, les résultats conduisent à proposer une révision du tracé de la dérive apparente des pôles pour l'intervalle de ~2300–~1650 Ma en Amérique du Nord.

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## Introduction

The Cobalt Plate (Fig. 1) of the Southern Structural Province is an ~3 km thick veneer of slightly deformed ~2.3 Ga Proterozoic strata resting unconformably on Archean rocks of the Superior Structural Province (Roscoe 1973). The Grenville Front boundary fault (Lumbers 1971) forms the southern boundary of the Cobalt Plate, separating it from the high-grade metamorphic terrain of the Grenville Structural Province. This terrain was metamorphosed during the Grenville Orogeny ~1000 Ma ago. Rock units in the Cobalt Plate and elsewhere in the Southern Province have been extensively studied paleomagnetically, leading to the identification of metamorphic remanence components in these rocks. This paleomagnetic study concentrates on rock units immediately north of the Grenville Front near River Valley, Ontario in order to determine the degree and penetration of the Grenvillian remanence remagnetization into the Cobalt Plate.

## Geology

The early Aphebian Proterozoic sediments of the Cobalt Plate belong to the Huronian Supergroup. In the study area, the Gowganda Formation is the only

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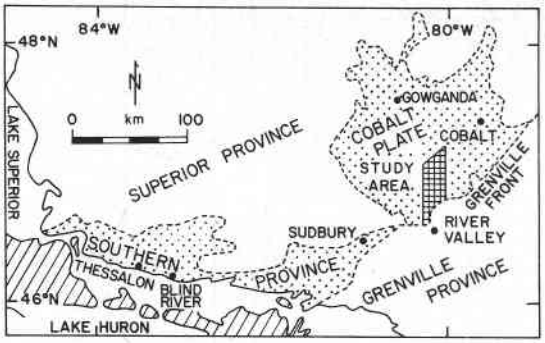



FIG. 1. Location of the study area relative to the Grenville Front in the North Shore area of Ontario.

Huronian Supergroup unit to outcrop extensively. Small areas of the underlying Mississagi and overlying Lorrain clastic sediments are present. The Gowganda Formation comprises polymictic conglomerate, quartz arenite, arkose, and argillite and has a maximum thickness of 1500 m (Roscoe 1973). It gives a concordant early Aphebian Rb–Sr whole-rock isochron age of 2290 ± 90 Ma (Fairbairn *et al.* 1969).

The Gowganda Formation has been extensively intruded by Nipissing sills in the study area. The sills are composed of subophitic gabbro, diabase, and late quartz monzonite differentiates. Some of these irregular sills are hundreds of metres thick (Dressler 1979). As determined by Rb–Sr whole-rock isochrons (Van Schmus 1965; Fairbairn *et al.* 1969) and K–Ar ages (Lowden *et al.* 1963), the sills were emplaced about 2165 ± 25 Ma ago during the late stages of the Penokean Orogeny (Church 1968).

The Proterozoic strata are nearly horizontal and are regionally metamorphosed to the lowest greenschist facies (chlorite subfacies) (Card and Pattison 1973) except near the Grenville Front (Dressler 1979). Minor deformation, yielding dip angles of a few degrees, occurred during the Penokean Orogeny before and during Nipissing emplacement (Symons 1970; Symons and Londry 1975). Within ~3 km of the front, these strata become isoclinally folded, exposing limited outcrop areas of the older Mississagi Formation. The metamorphic grade increases progressively through the biotite–chlorite and biotite–garnet facies towards the front. At the front all rock units commonly exhibit cataclastic texture from dynamic metamorphism (Lumbers 1971).

South of the front in the Grenville Province medium- to high-grade metamorphic rocks of the amphibolite facies outcrop. These rocks were metamorphosed during the Grenville Orogeny. As thermochron analysis of K–Ar radiometric data shows, the orogeny reset the isotope decay systems in the River Valley area to give ages between 950 and 1000 Ma (Harper 1967; Baer 1976). Brocoum and Dalziel (1974) summarized the substantial evidence that these biotite–plagioclase–garnet gneisses and migmatites were originally Proterozoic strata. They suggested that the gneisses and migmatites are metamorphosed Gowganda graywackes and arkoses. Similarly the amphibolites are derived from Nipissing diabase. The gneisses and migmatites have been intruded by large anorthosite plutons. These plutons show less evidence of strong tectonic deformations and locally retain massive zones and primary layering, and hence were probably emplaced late in the Grenville Orogeny (Lumbers 1971).

## Sampling

The Nipissing diabase and Gowganda sediments were sampled at 88 sites (~500 cores, ~1000 specimens) along two profiles (Fig. 2) running north from the Grenville Front into the Cobalt Plate. The eastern profile starts just south of Lake Temagami and extends ~40 km close to the northern end of the lake, and uses mostly shoreline outcrops. The western profile starts 10 km northwest of River Valley and extends ~45 km northward along Highway 805 to Emerald Lake and beyond on logging roads. Five cores were drilled at each site and oriented *in situ* to within ~2° using a solar compass checked with a Brunton compass. At six Gowganda conglomerate sites, five cores were drilled in the matrix sediment and five cores in the clasts for use in a conglomerate test for remanence remagnetization (Graham 1949). Two or more right-cylindrical specimens of 2.5 cm diameter were cut from each core.

## Remanence measurements

### *AF demagnetization*

The natural remanent magnetization (NRM) of each specimen was measured on a Schonstedt SSM-1A spinner magnetometer. One specimen with a typical NRM intensity and direction was selected from each site and subjected to alternating-field (AF) step demagnetization using a Schonstedt AF demagnetizer. The steps were 5, 10, 15, 20, 30, 40, 50, 65, 80, and 100 mT. Vector subtraction was used to look for multiple components in individual specimens. Following the stability index method (Symons and Stupavsky 1974), an optimum AF demagnetization field was selected for cleaning the remaining specimens from the site. Reliable remanence data were selected by screening at the core and site levels. Only homogeneously magnetized cores that had vector resultant lengths (Fisher 1953) of $R > 1.90$ were accepted. The only homogeneously magnetized sites accepted were those having three or more accepted cores and a radius of the 95% confidence circle ($\alpha_{95}$) of $<15°$ for Nipissing diabase and $<20°$ for the Gowganda sediments (Table 1). Site means were aver-

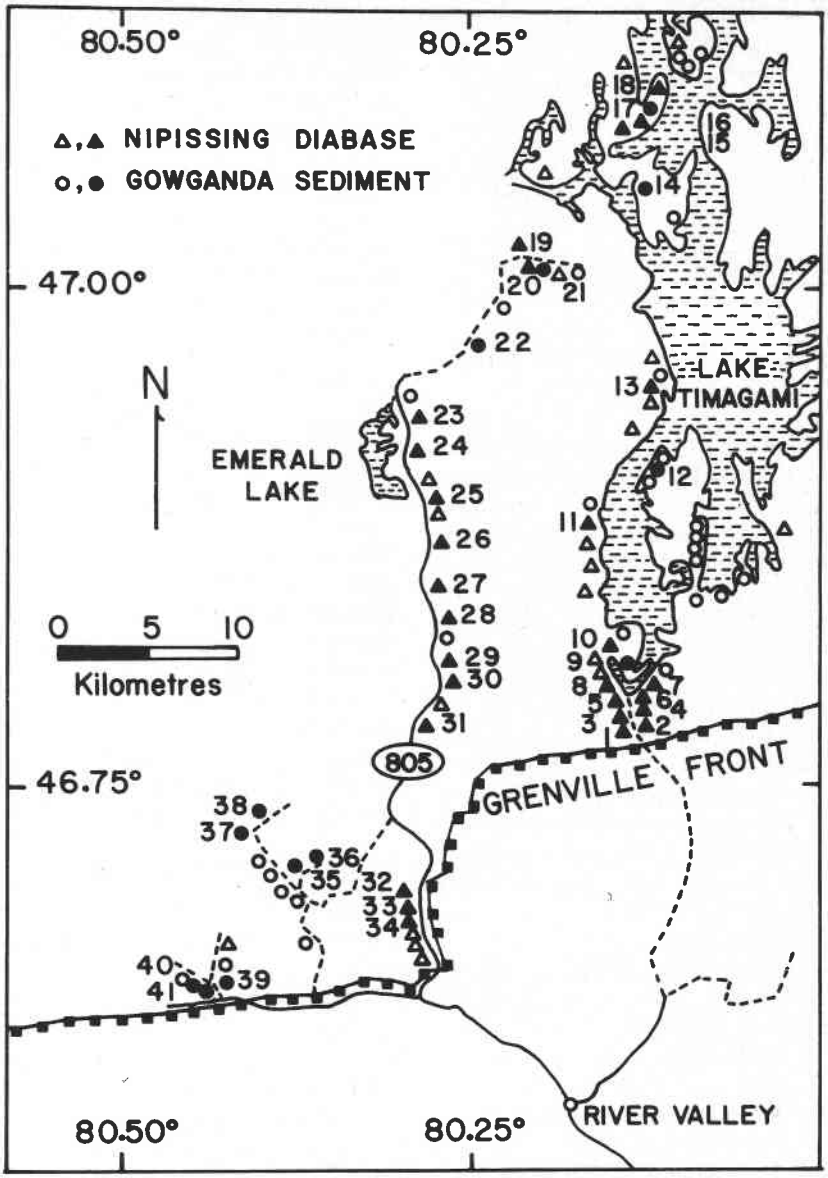



FIG. 2. Location of the sampled sites. Solid symbols (open) denote homogeneously magnetized Nipissing diabase (triangles) and Gowganda sediment (circles) sites, respectively, that were accepted (not accepted) for interpretation.

aged to get unit mean directions (Table 2; Fig. 3). AF step demagnetization results for Gowganda and Nipissing rocks have been discussed extensively previously (e.g., Symons 1967, 1970, 1971, 1975) and hence are discussed summarily here.

*Thermal demagnetization*

Sixteen untreated specimens were thermally demagnetized in seven steps up to 650°C in a Schonstedt oven. From their results it was decided to clean further the eight specimens from the 43 accepted AF demagnetized sites at temperatures of 550, 570, 620, and 650°C. The site mean directions were calculated from homogeneous specimen directions having a within-specimen angular standard deviation of $\leq 10°$ for 550 and 570°C, and of $\leq 20°$ for 620 and 650°C. The site mean directions with an $\alpha_{95} \leq 20°$ were used to determine the unit mean direction. The results are summarized in Table 3. Results from the thermal step demagnetization of Gowganda and Nipissing rocks have been discussed in detail in previous studies (e.g., Roy and Lapointe 1976; Morris 1977, 1979).

TABLE 1. Paleomagnetic data by sites

| Site | No. of accepted cores | AF demagnetizing field (mT) | Length of vector $R$ | $D$ (°) | $I$ (°) | Radius $\alpha_{95}$(°) |
|---|---|---|---|---|---|---|
| | | Gowganda sediments | | | | |
| 9 | 5 | 15 | 4.8771 | 14 | 81 | 14 |
| 12 | 4 | 10 | 3.9790 | 318 | 82 | 8 |
| 14 | 4 | 40 | 3.8617 | 336 | 50 | 20 |
| 17 | 5 | 20 | 4.9146 | 272 | 78 | 11 |
| 21 | 3 | 20 | 2.9639 | 307 | 76 | 17 |
| 22 | 4 | 15 | 3.9254 | 267 | 77 | 14 |
| 37 | 5 | 10 | 4.9419 | 39 | 85 | 9 |
| 38 | 5 | 10 | 4.9596 | 359 | 72 | 8 |
| | | Nipissing diabase | | | | |
| 1 | 5 | 80 | 4.8635 | 183 | 21 | 14 |
| 2 | 5 | 20 | 4.9378 | 169 | 44 | 10 |
| 3 | 5 | 40 | 4.8911 | 176 | 54 | 13 |
| 4 | 5 | 10 | 4.9727 | 181 | 44 | 6 |
| 5 | 3 | 20 | 2.9751 | 173 | 40 | 14 |
| 6 | 5 | 20 | 4.9980 | 186 | 42 | 2 |
| 7 | 5 | 20 | 4.9914 | 180 | 41 | 4 |
| 8 | 5 | 30 | 4.8981 | 171 | 42 | 12 |
| 10 | 3 | 20 | 2.9766 | 187 | 56 | 13 |
| 11 | 3 | 40 | 2.9778 | 183 | 63 | 13 |
| 13 | 3 | 65 | 2.8470 | 200 | 23 | 15 |
| 18[a] | 3 | 80 | 2.9847 | 350 | −40 | 11 |
| 19 | 4 | 20 | 3.9663 | 168 | 57 | 10 |
| 20 | 4 | 80 | 3.9616 | 153 | 51 | 11 |
| 23 | 5 | 50 | 4.9889 | 200 | 58 | 4 |
| 24 | 5 | 30 | 4.9527 | 215 | 13 | 8 |
| 25 | 3 | 30 | 2.8480 | 178 | 66 | 15 |
| 26 | 5 | 20 | 4.9962 | 204 | 50 | 2 |
| 27 | 5 | 30 | 4.9856 | 213 | 36 | 5 |
| 28 | 5 | 80 | 4.9592 | 214 | 30 | 8 |
| 30 | 5 | 40 | 4.9560 | 205 | 42 | 8 |
| 31 | 3 | 30 | 2.9220 | 152 | 23 | 15 |
| | | Nipissing remagnetization of Gowganda sediments | | | | |
| 35 | 4[b] | 30 | 3.9020 | 42 | −41 | 17 |
| 36 | 4 | 40 | 3.9800 | 29 | −24 | 8 |
| | | Sudbury dikes and remagnetization of Nipissing diabase | | | | |
| 15 | 4 | 10 | 3.9820 | 267 | 17 | 7 |
| 16 | 5 | 10 | 4.9746 | 270 | 18 | 6 |
| 29 | 5 | 20 | 4.9915 | 264 | 4 | 4 |
| | | Grenville remagnetization of Gowganda sediments | | | | |
| 39, matrix | 5 | 15 | 4.8828 | 82 | 80 | 13 |
| 39, clasts | 5 | 30 | 4.7798 | 101 | 77 | 18 |
| 40, clasts | 4 | 10 | 3.9412 | 73 | 73 | 13 |
| 40, matrix | 6 | 20 | 5.7118 | 197 | 83 | 16 |
| 41, clasts | 4 | 30 | 3.8480 | 94 | 53 | 20 |
| | | Grenville remagnetization of Nipissing diabase | | | | |
| 32, diabase | 4 | 15 | 3.8520 | 330 | 77 | 20 |
| 33, diabase | 4 | 15 | 3.8985 | 23 | 73 | 17 |
| 34, diabase | 4 | 10 | 3.9105 | 45 | 67 | 16 |

[a]Direction reversed to antiparallel to compute unit means.
[b]Site includes two Nipissing diabase and two remagnetized "baked Gowganda sediment" cores.

TABLE 2. Unit mean remanence directions after AF demagnetization and virtual geomagnetic poles

| | No. of sites | Resultant vector length $R$ | $D$ (°) | $I$ (°) | Precision parameter $k$ | Radius $\alpha_{95}$(°) | Long. (°W) | Lat. (°N) | Semi-axes of oval $P_{0.05}$ $d_p$(°) | $d_m$(°) |
|---|---|---|---|---|---|---|---|---|---|---|
| Gowganda sediments | | | | | | | | | | |
| Uncorrected[a] | 8 | 7.7815 | 327 | 78 | 32 | 10 | 109 | 63 | 10 | 19 |
| Tilt corrected | 8 | 7.2683 | 319 | 69 | 10 | 19 | | | | |
| Nipissing diabase | | | | | | | | | | |
| Uncorrected | 22 | 20.7613 | 186 | 44 | 17 | 8 | 85 | −17 | 6 | 10 |
| Uncorrected[b] | 11 | 10.4503 | 189 | 45 | 18 | 11 | | | | |
| Tilt corrected | 11 | 10.0794 | 192 | 55 | 11 | 15 | | | | |
| Nipissing remagnetization of Gowganda sediments | | | | | | | | | | |
| Uncorrected | 2 | 1.9680 | 35 | −33 | 31 | | 115 | −18 | | |
| Sudbury dikes and remagnetization of Nipissing diabase | | | | | | | | | | |
| Uncorrected[c] | 3 | 2.9766 | 267 | 13 | 85 | 13 | 164 | 3 | 7 | 14 |
| Tilt corrected | 3 | 2.9557 | 268 | 1 | 45 | 19 | | | | |
| Grenville remagnetization of: | | | | | | | | | | |
| (a) Gowganda sediments and clasts | | | | | | | | | | |
| Uncorrected[c,d] | 5 | 4.8675 | 94 | 75 | 30 | 14 | | | | |
| Tilt corrected | 5 | 4.8842 | 149 | 18 | 34 | 13 | | | | |
| (b) Nipissing diabase | | | | | | | | | | |
| Uncorrected[d] | 3 | 2.9575 | 21 | 74 | 47 | 18 | | | | |
| (c) Combined | | | | | | | | | | |
| Uncorrected | 8 | 7.7345 | 67 | 78 | 26 | 11 | 45 | 51 | 19 | 21 |

NOTES: Fold test: [a]significant at >95% confidence; [b]significant at >85% confidence; and [c]not significant. Conglomerate test: [d]significant.

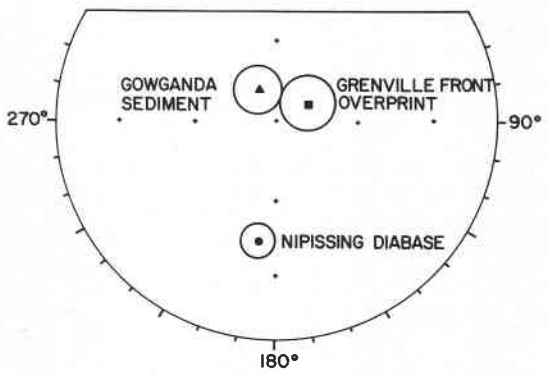



FIG. 3. Unit mean remanence directions shown on a Schmidt equal-area stereoplot with solid symbols denoting down directions. The means are circumscribed by their cones of 95% confidence.

## Results

*Gowganda sediments*

Eight sites of Gowganda sediments accepted after AF demagnetization are located >5 km north of the Grenville Front (Table 1). These eight site means were isolated in the 15 ± 5 mT range, except for one site. The loss of remanence coherence of the remaining 29 sites in the Gowganda sediments by 10 mT, and the AF step demagnetization data, indicate that their remanence is held dominantly in low AF coercivity domains. This is further emphasized by the thermal step demagnetization and bulk cleaning thermal data for the eight sites in which only three sites survive at 550°C and two sites at 570°C. The consequence is that the remanence of these sediments was blocked at relatively low temperatures and hence was very susceptible to remagnetization by a relatively mild thermal metamorphic event. This interpretation is consistent with the fold test of Graham (1949), which indicates with >95% confidence that the remanence of the eight AF sites is postfolding. Before tilt correction, these sites give a mean virtual geomagnetic pole (VGP) at 109°W, 63°N ($d_p = 10°$, $d_m = 19°$) (pole Gs5 in Fig. 4; Table 2).

The determined Gowganda pole is close to others reported for the Gowganda sediments near Elliot Lake at 123°W, 67°N (Gs2) (Symons 1975), near Cobalt at 158°E, 67°N (Gs3) (Roy and Lapointe 1976), and throughout the Southern Province at 151°W, 74°N (Gs4) (Morris 1977). Fold test by Symons (1975) and Morris (1977) indicate that this remanence is of postfolding diagenetic or Penokian metamorphic origin. Contact tests by Symons (1967), Roy and Lapointe (1976), and Morris (1979) indicate that this remanence predates Nipissing intrusion. Morris (1977) isolates a second

TABLE 3. Unit mean remanence directions on thermal demagnetization before tilt correction

| | Temp. (°C) | No. of sites | Resultant vector length $R$ | $D$ (°) | $I$ (°) | Radius $\alpha_{95}$(°) |
|---|---|---|---|---|---|---|
| Gowganda sediments | 550 | 3 | 2.9952 | 115 | 79 | 19 |
| | 570 | 2 | 1.9948 | 75 | 74 | 18 |
| Nipissing diabase | 550 | 19 | 17.6835 | 187 | 39 | 9 |
| | 570 | 19 | 17.3534 | 180 | 39 | 11 |
| | 620 | 12 | 11.1388 | 184 | 38 | 13 |
| | 650 | 6 | 5.6397 | 200 | 33 | 14 |
| Nipissing remagnetization of Gowganda sediments | 550 | 2 | 1.9992 | 38 | −45 | 7 |
| | 570 | 2 | 1.9958 | 38 | −47 | 16 |
| | 620 | 2 | 1.9785 | 35 | −48 | 37 |
| Sudbury dikes and remagnetization of Nipissing diabase | 550 | 3 | 2.9804 | 264 | −4 | 12 |
| | 570 | 3 | 2.9564 | 263 | 1 | 18 |
| Grenville remagnetization of Gowganda sediments and clasts | 550 | 5 | 4.8081 | 111 | 74 | 17 |
| | 570 | 4 | 3.8286 | 129 | 72 | 23 |
| | 620 | 2 | 1.9936 | 98 | 66 | 10 |

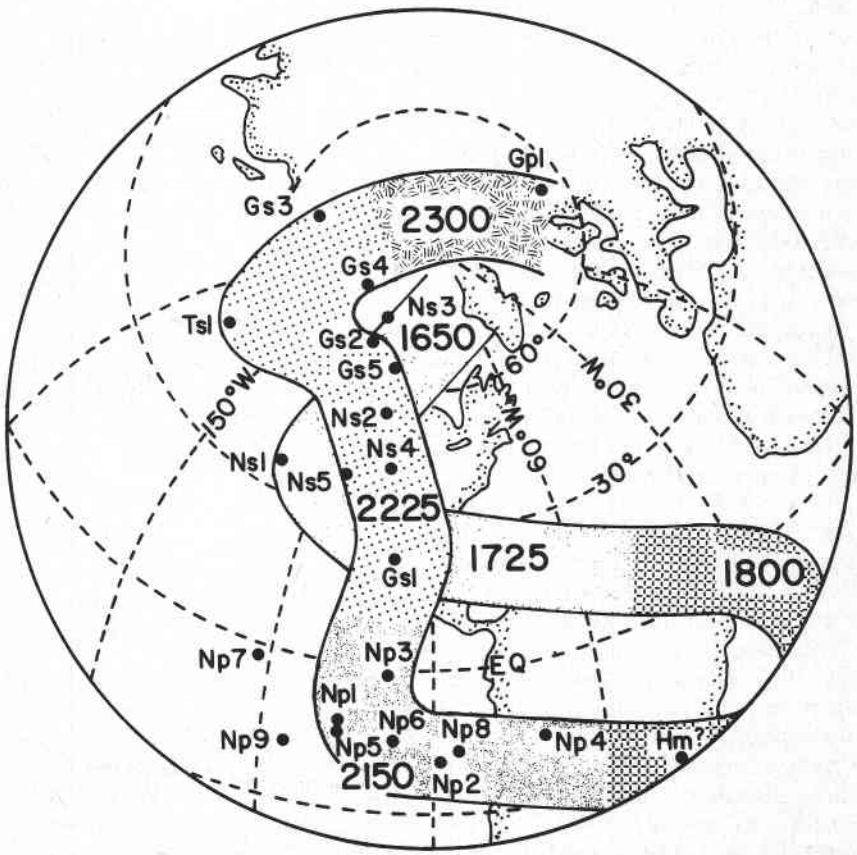



FIG. 4. Proposed APW path based on Southern Province paleomagnetic rock studies for time span 2300–1650 Ma. The poles are labelled as listed in the text.

component giving a pole at 35°E, 66°N (Gp1), which is prefolding (based on a fold test). The Gowganda sediment pole near the Silverfields mine, Cobalt at 97°W, 22°N (Gs1) reported by Symons (1967) comes from one 25 m section and may not be as reliable as the others. The Thessalon volcanics from the base of the Huronian Supergroup give a similar pole at 169°W, 47°N (Ts1) (Symons and O'Leary 1978). A fold test applied to these gently deformed volcanics was inconclusive.

*Nipissing diabase*

Twenty-two sites accepted after AF demagnetization are located >1 km away from the Grenville Front, in Nipissing diabase of the lowest greenschist facies (Card and Pattison 1973). Their site mean remanence directions form a coherent "south and down" population before tilt correction with a unit mean direction of 186°, 44° ($\alpha_{95}$ = 8°). The remanence direction of the most northerly site (18) in the sampling profiles is 350°, −40° and it has been reversed to compute the unit mean. AF step demagnetization of the pilot specimens for these sites (Fig. 5) indicates that the isolated remanence is characterized by a highly stable end point. Further, thermal step demagnetization shows that this direction is held up to 570°C in all specimens, and to 650°C in some specimens. When thermal cleaning is applied to all specimens, this direction is held to 570°C in 19 sites (Table 3), indicating a dominant magnetite carrier, and to 650°C in six sites, indicating a stable hematite carrier. This direction is antiparallel to the main N1 population of Nipissing directions (Morris 1979) found elsewhere in Ontario, and using the AF data it gives a VGP of 85°W, 17°S ($d_p$ = 6°, $d_m$ = 10°) (Np8) (Fig. 6). Similar VGP's based on "north and up" remanence directions from Nipissing sills have been reported from the Silverfields mine near Cobalt at 106°W, 12°S (Np1) (Symons 1967), from the Cobalt area at 88°W, 19°S (Np2) (Symons 1970), from the Gowganda area at 106°W, 14°S (Np5) (Symons and Londry 1975), and from the Cobalt area at 96°W, 15°S (Np6) (Roy and Lapointe 1976). Two VGP's for the Nipissing diabase sills near Blind River at 98°W, 1°S (Np3) (Symons 1971) and at 70°W, 16°S (Np4) (Patel and Palmer 1974) are based on four and five sites, respectively, having "south and down" remanence directions. Contact tests reported by Symons (1967) and Morris (1979) confirm that this is a primary intrusive direction. Both tests provide profiles that include the sill and baked contact with one direction and the host rock with a different and consistent Gowganda direction. Morris (1979) also reported a "south and down" pole of 122°W, 3°S (Np7) based on two sill sites and two baked contact zone sites from a restricted locality near Elliot Lake.

The Nipissing diabase yields a second less common

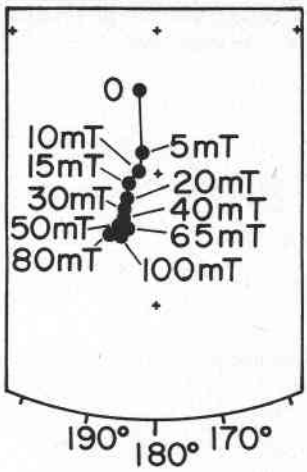



FIG. 5. Mean AF step demagnetization directions for 22 Nipissing diabase specimens from the accepted sites. The mean direction at each demagnetizing field is plotted on a segment of an equal-area stereonet. The circle of 95% confidence decreases from an NRM value of 19.6° to a minimum of 9.8°C at a field of 65 mT and then increases, reaching 13.3° at 100 mT.

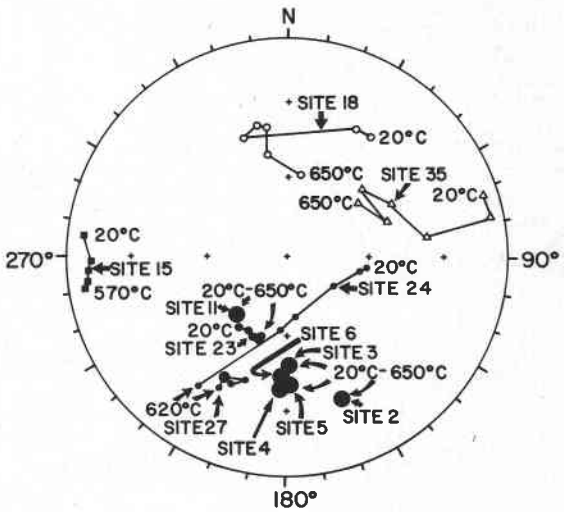



FIG. 6. Examples of change in remanence direction after thermal step demagnetization for specimens of Nipissing diabase (circles), baked Gowganda sediment (triangles), and Sudbury olivine diabase (squares).

remanence direction, which Morris (1979) designated as N2. It is a steeply inclined "northwest and down" direction. It is defined by VGP's from Elliot Lake dikes at 131°W, 36°N (Ns1) (Symons 1971), from Elliot Lake sills at 106°W, 53°N (Ns2) (Patel 1971), from Gowganda at 122°W, 73°N (Ns3) (Symons and Londry 1975), and from Cobalt at 102°W, 42°N (Ns4) (Roy and Lapointe 1976). Roy and Lapointe (1976) also reported a contact test in which the baked zone gives a pole of

114°W, 39°N (Ns5), showing that the N2 direction must also be primary. This test is less reliable because it did not include unaltered Firstbrook (upper Gowganda) host rock remanence directions in the profile. Also, Symons *et al.* (1976) reported that the Firstbrook section on Highway 11 near Cobalt contains three normal zones with directions of 325°, 58° and three reversed zones with directions of 288°, −56°. Symons (1971) observed that the N2 dikes cross-cut the N1 sills near Elliot Lake, which means that the Nipissing was emplaced in two phases with N1 predating N2. Both N1 and N2 are found in the sill at Gowganda (Symons and Londry 1975). The spatial distribution of sites in the sill places all the N1 sites in the southwest side with one exception, thereby allowing the possibility of two separate phases of emplacement. Fold tests on the N1 directions of the Nipissing sills at Cobalt (Symons 1971) and Gowganda (Symons and Londry 1975) show with 95% confidence that the gentle basin and arch structure in the sills is primary, and therefore that they were emplaced in gently folded Huronian rocks. Thus the Gowganda strata were folded during the Penokean Orogeny. A fold test applied to the 11 of 22 sites in this study for which very reliable attitudes could be obtained from the maps indicates with 85% confidence that their remanence is also postfolding.

Three apparent "Nipissing diabase" sites (15, 16, 29, Table 1) appear to form a second AF population having a mean direction of 267°, 13° ($\alpha_{95} = 13°$), giving a pole of 164°W, 3°N. Thermal cleaning of these three sites at 550 and 570°C continues to further isolate this direction to 263°, 1° at 570°C, indicating that it is held by highly stable magnetite carriers. This direction is very similar to those given by the 1250 Ma Sudbury olivine diabase dikes (268°, 1° and 265°, 2° reported by Larochelle (1967) and Palmer *et al.* (1977), respectively). Thin-section examination of specimens from these three sites shows that sites 15 and 16 are in Sudbury olivine dikes and that site 29 is in highly altered Nipissing diabase that may have been baked by a known adjacent Sudbury dike (Fig. 2; Ontario Geological Survey map 2361). Also the pole of 164°W, 3°N is fairly close to that of 180°W, 9°S (Roy and Lapointe 1976) for Nipissing diabase cut by late faults near Cobalt. The southeast faults could be related to the same tectonic event as the intrusion of the Sudbury dikes.

*Nipissing remagnetization of Gowganda sediments*

Site 36 is in sediment close to a large Nipissing sill, and site 35 straddles the Nipissing–Gowganda contact with two accepted cores in the diabase and two in baked sediment. The sediment has clearly been baked by the sill and gives the same Nipissing remanence direction. This is not a complete contact test because the sampling profile does not extend into unaltered host rock. However, when combined with data from surrounding unaltered Gowganda sediment sites, this is a baked contact test that indicates that the Nipissing N1 direction with a pole at 115°W, 18°S (Np9) is primary.

*Revision to apparent polar wander path*

The various poles for the Huronian Supergroup and Nipissing studies are plotted in Fig. 4. They suggest a revision in the apparent polar wander (APW) path put forward by Irving (1979), which Morris (1979) criticized. In essence, these poles do not change appreciably the position of the path, but rather shift the time positions farther down the path so that the N1 poles fall at ~2150 Ma instead of ~1825 Ma and the N2 poles on the return early Paleohelikian segment fall at ~1650 Ma instead of at ~2150 Ma. This revised path fits the other time calibration poles used by Irving (1979) rather well. For example, the Abitibi dikes fall on the 2200 Ma portion rather than the 2050 Ma portion, which better fits their known radiometric age data. Symons *et al.* (1981) noted that Abitibi dikes with equatorial N1 poles give a Rb–Sr age of 2206 ± 132 Ma, whereas those with high-latitude N2 poles give an age of 2102 ± 67 Ma; they also noted that a positive contact test is present for both N1 and N2 dikes. The poles for the Indian Harbour dikes, the Indin dikes, the Otish gabbro, and the Owl Creek dikes are other examples of better fits. The Molson dike and the Big Spruce Complex fit both paths equally well, depending on which components are examined in the latter case. The pole for the Wind River dikes gives a poorer fit. The dates for other poles such as those of the Gunflint Formation and the Animikie hematite ore are not tightly enough constrained to indicate which path provides a better fit. Poles from the Labrador Trough such as those of the Castignon Complex, Menihek Formation, Seward subgroup, and Sokoman Formation all give a poorer fit. These rocks give a date of ~1870 Ma, which fits the Irving (1979) path; however, the rocks have also been "intensely folded and faulted with overthrusting" during the Hudsonian Orogeny (Park 1977), which means substantial geotectonic rotation may have occurred. Units from the Great Slave Supergroup in the Coronation geosyncline give scattered poles with tectonic problems; these poles may fit either the Irving (1979) path or the proposed path following the earlier McGlynn and Irving (1978) interpretation. Summarizing, on balance the proposed path appears to provide a better fit to the available data although neither path is particularly reliable.

*Grenvillian remanence north of the front*

Evidence of significant remagnetization by Grenvillian metamorphism is found in eight sites that are located within ~2 km of the front: two sites (39, 40) are located in Gowganda conglomerate matrix; three sites (39, 40, 41) are in highly sheared and deformed

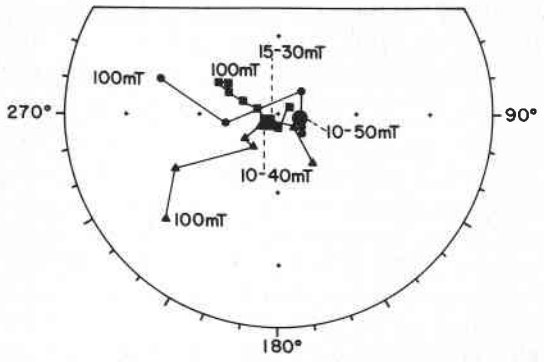



FIG. 7. Remanence directions of three Gowganda clast specimens showing the isolation of a Grenville Orogeny remagnetization during AF step demagnetization in the ~10–~50 mT range.

Gowganda conglomerate clasts; and three sites are in metamorphosed Nipissing diabase of the amphibolite facies. These eight sites form a coherent population with a mean remanence direction of 67°, 78° ($\alpha_{95}$ = 11°). Thirteen of the 15 clast cores give acceptable mean remanence directions after AF demagnetization in 20–30 mT fields, and three of the clast pilot specimens retained the stable direction to at least 40 mT (Fig. 7). The grouping of these fairly stable clast remanence directions provides a conglomerate test that indicates they have been substantially remagnetized by a metamorphic event. On thermal demagnetization, the amphibolitized Nipissing diabase did not retain a coherent direction at 550°C or above. However, the Gowganda sediments and clasts did retain the Grenville direction in all five sites at 550°C, in four sites at 570°C, and in two sites at 620°C. A fold test using the five Gowganda sites with known attitudes is inconclusive because it results in little change in the degree of scatter, and therefore either a postfolding or prefolding origin for this remanence component may be valid. If it is postfolding, then the VGP for all eight sites is 45°W, 51°N, ($d_p$ = 19°, $d_m$ = 21°). This pole falls on the 1000 Ma portion of the APW curve for the Grenville (McWilliams and Dunlop 1978; Irving 1979), which is consistent with the preferred interpretation that it is a Grenville orogenic remagnetization component. If the remanence is prefolding, then the five Gowganda sites give a tilt corrected VGP of 46°W, 28°S (Hm?), which is well removed from the Grenville APW path but falls on the 1950 Ma portion of the late Aphebian APW path (Fig. 4). This position is consistent with the possible interpretation that it is a Hudsonian metamorphic component. Both interpretations support the conclusion of Brocoum and Dalziel (1974) that the intense deformation near the Grenville Front predates the Grenville Orogeny, and they further imply that the deformation predates either 1000 or 1950 Ma, respectively.

The Grenvillian metamorphic remagnetization in the River Valley area of the Southern Province is limited to an ~2 km wide zone of highly metamorphosed rocks near the Grenville Front. In this zone the Gowganda conglomerate clasts, matrix, and metamorphosed Nipissing diabase carry a stable remanence of Grenvillian age. Outside this zone, no significant Grenville remanence component survives AF cleaning in either the Nipissing diabase or the Gowganda sediments. For example, unmetamorphosed Nipissing diabase at sites 1–4 (Table 1), which are located only a few kilometres north of the front, shows no evidence of Grenvillian remanence components. Palmer *et al.* (1977) completed a paleomagnetic study on the ~1250 Ma Sudbury olivine diabase dikes in the Southern Province, some of which cross the Grenville Front and continue into the Grenville Province. They found that significant Grenvillian remanence components in these dikes occurred only within ~1 km of the front in the Southern Province. Their results also conflict with the interpretation that the Grenville remagnetization predates 1250 Ma.

## Conclusions

Several conclusions are drawn from this paleomagnetic study.

(1) Significant remagnetization of the primary remanence of the Huronian Gowganda sediments and Nipissing diabase of the Southern Province by the Grenville Orogeny is confined to a narrow zone extending ~2 km north of the Grenville Front.

(2) Significant folding deformation at the Grenville Front predates the Grenville Orogeny, as has been suggested by Brocoum and Dalziel (1974). This conclusion is drawn from the fact that eight sites of Nipissing diabase, Gowganda sediments, and conglomerate clasts in the Gowganda sediments give a combined postfolding remanence with a Grenvillian pole at 45°W, 51°N ($d_p$ = 190°, $d_m$ = 21°).

(3) The 22 sites of Nipissing diabase sills north of the zone give a prefolding primary remanence direction with an N1 pole at 85°W, 17°S ($d_p$ = 6°, $d_m$ = 10°). Except for the most northerly site (18) and the baked contact site (35), all sites give a normal "south and down" remanence direction. Conversely, except for previously reported Nipissing studies relatively near the front at Blind River, all other studies giving N1 poles have been based on antiparallel "north and up" remanence directions. The geographic distribution of "south and down" directions near the front and "north and up" directions in the northern Southern Province implies that either the Nipissing diabase sills were emplaced as two distinctly different events or the paleogeomagnetic field changed polarity at least once during the intrusion event. In the former case, the antiparallelism of the directions suggests that the two events were only a few million

years apart. The latter case suggests that emplacement was sequential from south to north or north to south across the Southern Province. Finally, the antiparallelism provides additional evidence that the N1 poles are primary.

(4) Three sites of apparent Nipissing diabase give a pole at 164°W, 3°N ($d_p = 7°$, $d_m = 14°$), which is similar to that found for the Sudbury olivine diabase dikes. Thin-section observations confirm that two of the sites are in olivine diabase. The third site is adjacent to the contact of a known wide diabase dike and is thought to have been baked by it.

(5) The eight accepted sites of Gowganda sediment north of the ~2 km wide zone at the front show no evidence of a Grenvillian metamorphic remagnetization. Their remanence is postfolding at ~95% confidence and their pole position of 109°W, 63°N ($d_p = 10°$, $d_m = 19°$) is similar to those found elsewhere in the Southern Province. Their remanence appears to have been acquired during the Penokean Orogeny, as suggested by Morris (1979).

(6) The results lead to a modification of the APW path proposed by Irving (1979) for the 2300–1650 Ma time interval for North America.

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